

A Multi-Layered Safety Controller for Automated Anesthesia

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Abstract: This paper examines the problem of controlling the automated clinical administration of propofol, an anesthetic drug, to ensure patient safety. Existing anesthesia controllers typically adjust drug infusion rates to achieve a target patient sedation level. Our goal is to complement this previous research by developing an algorithm that (i) provides formal anesthesia safety guarantees and (ii) can be used either as a standalone controller or as a supervisory safety governor for existing controllers. Specifically, we introduce two barrier functions that quantify the safety implications of over- and under-dosing. We then construct novel nonlinear second-order barrier constraints that achieve safety in a “graceful” manner, by providing multi-layered safety guarantees, even though the dynamics of the underlying barrier functions have a relative degree of 2. Finally, we design a safety control algorithm that enforces these barrier constraints in the absence of full state measurements by combining the constraints with an interval observer. Simulation results that the proposed approach succeeds in enabling the safe induction and maintenance of target sedation levels, with reasonable induction times and “graceful” multi-layered safety assurances.

Keywords: Automated anesthesia, graceful safety control

1. INTRODUCTION

This paper examines the problem of ensuring patient safety when feedback control is used to administer anesthetic drugs such as propofol during surgery. Motivation for this work is twofold. First, compared to manual techniques like open-loop target-controlled infusion (TCI), automated control reduces clinician workload, tracks target hypnosis levels more accurately, and requires less cumulative drug administration (Khodaei et al., 2020; Pasin et al., 2017). These advantages are important for improving clinical outcomes and protecting patient safety. Second, the accurate control of propofol infusion is critical for patient safety. Under-dosage can lead to patient awareness, thereby triggering post-traumatic stress disorder (Avidan et al., 2008). On the other hand, as sedatives like propofol are linked to hypotension (Saugel et al., 2022; Dumont, 2012), over-dosage can cause organ failure and patient death. Anesthesia must therefore be tightly controlled to be neither insufficient nor excessive.

There is a rich literature on the closed-loop control of propofol infusion. Much of this literature represents drug infusion dynamics using compartmental pharmacokinetic-pharmacodynamic (PKPD) models (Derendorf and Meibohm, 1999). Many anesthesia control algorithms in the literature use proportional-integral-derivative (PID) control. PID gains can be tuned using either population data of infusion dynamics (Van Heusden et al., 2014), or patient-specific online adaptive tuning (Soltesz et al., 2013), or both. Optimization tools such as genetic algorithm can also be used to schedule different PID gains for distinct phases of anesthesia, such as *induction* ver-

sus *maintenance* (Padula et al., 2017). Beyond PID control, the literature also explores feedforward adaptive control (Nunes et al., 2009), model predictive control (Ionescu et al., 2008), and fuzzy logic learning (Wei et al., 2020). Ensuring patient safety is paramount regardless of which of these anesthesia control algorithms is in use.

One limitation of the literature is that while existing anesthesia controllers typically achieve good sedation performance, most do not augment this performance with formal safety guarantees. This creates a need for an additional control layer serving as a “safety governor” for lower-level anesthesia controllers. The goal of this paper is to address this research gap through three novel contributions:

- First, we develop an anesthesia control algorithm that provides formal guarantees of safety against both under- and over-dosage. The algorithm complements the existing literature in the sense that it can be used either as a standalone anesthesia controller (as shown in this paper) or as a supervisory governor that adds safety guarantees to existing controllers from the literature. We design this algorithm for a PKPD model of propofol infusion from the literature (Eleveld et al., 2018), but the underlying design approach can be extended to other anesthetic drugs.
- Second, we design the above controller using a novel approach that builds on our previous safety control research (Moon et al., 2025; Yin et al., 2025; Moon, 2025). This approach is similar to classical control barrier function (CBF) methods in how it (i) uses barrier functions and (ii) constrains their dynamics to quantify and ensure safety, respectively (Ames et al.,

2017). However, it has the advantage of using nonlinear second-order dynamic constraints to provide multi-layered safety guarantees, where for each safety hazard, a “primary” safety condition is sought whenever possible, and a “failsafe” condition is guaranteed otherwise. We achieve these multi-layered “graceful” safety guarantees even though the underlying barrier function dynamics have a relative degree of 2.

- Third, we combine the above graceful controller with an interval observer in order to provide safety guarantees using partial state measurements.

The remainder of this paper presents a PKPD model of propofol infusion dynamics (Section 2) plus the key ideas behind graceful safety control and interval observers (Section 3). The paper then presents the proposed controller and demonstrates its effectiveness in simulation (Section 4). Finally, Section 5 summarizes the paper’s conclusions.

2. PLANT MODEL

We use a three-compartment PKPD model of propofol infusion dynamics from the literature as the foundation of this work (Derendorf and Meibohm, 1999; Eleveld et al., 2018). The model’s state vector is given by $\mathbf{x}(t) = [c_1(t) \ c_2(t) \ c_3(t) \ c_e(t)]^T$, where the state variables $c_{1,2,3,e}(t)$ represent propofol concentrations ($\mu\text{g mL}^{-1}$) in the plasma, fast peripherals, slow peripherals, and effect site (i.e., brain), respectively. The dynamics of these propofol concentrations are governed by:

$$\begin{aligned}\dot{c}_1 &= k_{12}(c_2 - c_1) + k_{13}(c_3 - c_1) - k_{10}c_1 + u/V_1, \\ \dot{c}_2 &= k_{21}(c_1 - c_2), \quad \dot{c}_3 = k_{31}(c_1 - c_3), \\ \dot{c}_e &= k_{e0}(c_1 - c_e),\end{aligned}\quad (1)$$

where k_{ij} are coefficients of transport from compartment j to compartment i , k_{10} reflects metabolic elimination, k_{e0} describes transport from plasma to the brain, and V_1 is the effective volume of the plasma compartment in mL . The control input $u(t)$ is the rate of propofol infusion in $\mu\text{g min}^{-1}$. The above differential equations collectively constitute a “PK” submodel that describes compartmental dynamics. The “PD” submodel that describes how propofol affects the body is represented using a nonlinear static Hill function, as shown below:

$$\text{BIS}(c_e) = \text{BIS}_0 \left(1 - \frac{c_e^\gamma}{c_e^\gamma + c_{e,50}^\gamma}\right) \quad (2)$$

The above Hill function maps the effect site concentration, $c_e(t)$, to a *bispectral index* (BIS) value quantifying brain activity on a 0-100% scale. Patient safety during sedation depends strongly on BIS, with high and low BIS values indicating propofol under- and over-dosing, respectively. The parameters of the Hill equation are the patient’s baseline waking BIS, BIS_0 ; the Hill cooperativity constant, γ ; and the effect site concentration at 50% BIS, $c_{e,50}$.

In a typical clinical setting, the state variables $c_{1,2,3,e}(t)$ are not directly measured, but BIS measurements are available. Moreover, the Hill function is strictly monotonic in $c_e(t)$, and therefore invertible. This means that if the parameters of this Hill function are known, one can invert this function to estimate $c_e(t)$. This insight makes it possible to represent the above PKPD model using the linear and time-invariant (LTI) state-space representation

Parameter	Value	Units
k_{10}	0.2850	min^{-1}
k_{12}	0.2787	min^{-1}
k_{13}	0.1768	min^{-1}
k_{21}	0.0686	min^{-1}
k_{31}	0.0041	min^{-1}
k_{e0}	0.1460	min^{-1}
V_1	6280	mL
$c_{e,50}$	3.08	$\mu\text{g mL}^{-1}$
BIS_0	93	%

Table 1. Model Parameters

below, with the understanding that measurements of the output of this model may be affected by both sensor noise and errors in assumed Hill function parameters:

$$\begin{aligned}\dot{\mathbf{x}}(t) &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t), \quad y(t) = \mathbf{C}\mathbf{x}(t), \\ \mathbf{A} &= \begin{bmatrix} -k_{10} - k_{12} - k_{13} & k_{12} & k_{13} & 0 \\ k_{21} & -k_{21} & 0 & 0 \\ k_{31} & 0 & -k_{31} & 0 \\ k_{e0} & 0 & 0 & -k_{e0} \end{bmatrix}, \\ \mathbf{B} &= \begin{bmatrix} 1/V_1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{C} = [0 \ 0 \ 0 \ 1].\end{aligned}\quad (3)$$

Notice that this system has a relative degree of 2, as $u(t)$ appears in $\ddot{y}(t) = \ddot{c}_e(t)$ but not $\dot{y}(t)$. Table 1 lists the parameter values for a reference 35-year-old, 70 kg, 170 cm male, approximated using data from Eleveld et al. (2018).

3. APPROACH: GRACEFUL SAFETY CONTROL AND INTERVAL ESTIMATION

The goal of this paper is to develop an algorithm that ensures patient safety by enforcing clinician-selected upper and lower bounds on effect site concentration, $x_4(t) = c_e(t)$. This algorithm builds on two key topics from the literature, namely, graceful safety control and interval estimation. This section introduces these topics, based on the existing literature, through the following results.

3.1 Graceful Control

The core foundations of our proposed graceful safety controller are summarized by the following theorem.

Theorem 1: Consider a dynamic system governed by the state equation $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$, where $\mathbf{x}(t) \in \mathcal{R}^n$. Moreover, let $h_g : \mathcal{R}^n \rightarrow \mathcal{R}$ be a “graceful” control barrier function, and suppose that the control input $\mathbf{u}(t)$ is chosen at every instant in time, t , such that the dynamics of $h_g(\mathbf{x})$ satisfy the following inequality:

$$\ddot{h}_g + (\lambda_1 + \lambda_2)\dot{h}_g + \lambda_1\lambda_2 \frac{h_g}{1 + h_g} \geq 0, \quad (4)$$

for some positive real constants $\lambda_{1,2}$. Then the following two sets are forward-invariant:

$$S_{g1,2} = \{\mathbf{x} \in \mathcal{R}^n : \dot{h}_g(\mathbf{x}) + \lambda_{1,2}h_g(\mathbf{x}) \geq 0\} \quad (5)$$

Moreover, for any initial condition, $\mathbf{x}(0)$ that satisfies $-1 < h_g(\mathbf{x}(0)) < 0$, the trajectory of $h_g(\mathbf{x}(t))$ satisfies:

$$\lim_{t \rightarrow \infty} h_g(\mathbf{x}(t)) \geq 0 \quad (6)$$

The proof of the above theorem can be found in earlier work by the authors (Moon, 2025), and is omitted here

for brevity. However, for the purposes of this paper, the intuition behind it is crucial.

The function h_g is a “graceful” safety barrier function. Specifically, it is constructed such that it is positive when the underlying system is safe, negative when the system is unsafe, and equal to -1 when the system is unsafe to a catastrophic degree. Then Theorem 1 provides multiple layers of safety guarantees. Top-layer safety is provided by the forward invariance of two linear constraints on the relationship between the graceful barrier function, $h_g(\mathbf{x}(t))$, and its time derivative. Forward invariance means that if these linear constraints are satisfied at the beginning of time, they will be satisfied perpetually afterwards.

This “primary” safety guarantee is analogous to the safety assurances provided by classical exponential CBF methods from the literature (Moon, 2025; Ames et al., 2017). However, in addition to this primary assurance, graceful control provides an important secondary “failsafe” guarantee: even if the system is initialized in an unsafe but not catastrophic situation (i.e. $-1 < h_g(\mathbf{x}(0)) < 0$), it is guaranteed to return to safety eventually. Achieving this secondary guarantee is particularly valuable for biomedical applications, such as automated anesthesia. It is also important to note that Theorem 1 provides these guarantees even if the underlying dynamics of the graceful CBF, $h_g(\mathbf{x})$, have a relative degree of 2, as is the case in this paper’s PKPD model.

3.2 Interval Estimation

Theorem 1 provides a framework for *controlling* anesthesia to ensure safety, assuming state feedback control. In the absence of full state measurement, it is important to combine such graceful control with an *estimation* framework that provides equally rigorous guarantees. We pursue this goal using an *interval observer*, as discussed below.

An interval observer aims to estimate an interval, $[\underline{\mathbf{x}}(t), \bar{\mathbf{x}}(t)]$ with $\underline{\mathbf{x}}(t) \leq \bar{\mathbf{x}}(t)$ such that $\mathbf{x}(t) \in [\underline{\mathbf{x}}(t), \bar{\mathbf{x}}(t)]$ for all time. Suppose that the system of interest has the following single-input, single-output (SISO), linear, and time-invariant (LTI) dynamics:

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t) + \mathbf{d}(t), \quad y = \mathbf{C}\mathbf{x}(t) + v(t), \\ d(t) &\in [\underline{\mathbf{d}}(t), \bar{\mathbf{d}}(t)], \quad |v(t)| \leq V, \quad \forall t \end{aligned} \quad (7)$$

In the above model, $u(t)$ represents a known (and bounded) system input for all time. Moreover, $\mathbf{d}(t)$ represents a disturbance vector with known component-wise bounds $\underline{\mathbf{d}}(t), \bar{\mathbf{d}}(t)$ for all time. Finally, $v(t)$ represents a scalar sensor noise signal with known bounds $-V, V$ for all time. Given this model, Efimov and Raïssi (2016) propose the following interval observer:

$$\begin{aligned} \dot{\bar{\mathbf{x}}} &= \mathbf{A}\bar{\mathbf{x}} + \mathbf{B}u + \mathbf{L}(y - \mathbf{C}\bar{\mathbf{x}}) + |\mathbf{L}|V + \bar{\mathbf{d}}, \\ \dot{\underline{\mathbf{x}}} &= \mathbf{A}\underline{\mathbf{x}} + \mathbf{B}u + \mathbf{L}(y - \mathbf{C}\underline{\mathbf{x}}) - |\mathbf{L}|V + \underline{\mathbf{d}}, \end{aligned} \quad (8)$$

where $\mathbf{L}$ is an observer gain matrix, and $|\mathbf{L}|$ is its element-wise absolute value. Given this estimation law, one can define the following two estimation errors:

$$\bar{\mathbf{e}}(t) := \bar{\mathbf{x}}(t) - \mathbf{x}(t), \quad \underline{\mathbf{e}}(t) := \mathbf{x}(t) - \underline{\mathbf{x}}(t) \quad (9)$$

One important goal in interval observer design is to ensure that the above estimation errors remain element-wise nonnegative at all moments in time. This ensures that the

true state is contained within the estimated interval and that the interval is valid (i.e. $\underline{\mathbf{x}}(t) \leq \bar{\mathbf{x}}(t)$). The dynamics of these estimation errors are given by:

$$\begin{aligned} \dot{\bar{\mathbf{e}}} &= (\mathbf{A} - \mathbf{L}\mathbf{C})\bar{\mathbf{e}} + \mathbf{L}v + |\mathbf{L}|V + \bar{\mathbf{d}} - \mathbf{d}, \\ \dot{\underline{\mathbf{e}}} &= (\mathbf{A} - \mathbf{L}\mathbf{C})\underline{\mathbf{e}} - \mathbf{L}v + |\mathbf{L}|V + \mathbf{d} - \underline{\mathbf{d}} \end{aligned} \quad (10)$$

Now suppose that the initial interval estimate $[\underline{\mathbf{x}}_0, \bar{\mathbf{x}}_0]$ contains the true initial state $\mathbf{x}_0$. The quantities $\pm \mathbf{L}v(t) + |\mathbf{L}|V, \bar{\mathbf{d}}(t) - \mathbf{d}(t)$ and $\mathbf{d}(t) - \underline{\mathbf{d}}(t)$ are component-wise non-negative by construction. Therefore, the sole requirement for the estimated interval to contain the true state at all times is for the matrix $\mathbf{A} - \mathbf{L}\mathbf{C}$ to be Metzler for the errors’ nonnegativity, and Hurwitz for the errors’ asymptotic stability (Efimov and Raïssi, 2016). Any estimator that meets these conditions is an interval observer.

4. SIMULATION STUDIES

This section presents two simulation studies demonstrating the performance of the proposed graceful safety controller. The first study assumes the availability of full state measurements, whereas the second study employs the proposed interval estimator. Both studies use the nominal patient parameters from Table 1.

4.1 Graceful Control with Full State Measurement

We begin graceful safety control design by constructing two barrier functions representing the impacts of propofol under- and over-dosage, respectively. The first barrier function is defined as follows:

$$h_{g,1}(c_e) := \frac{c_e - c_{g,1}}{c_{g,1} - c_{e,min}} \quad (11)$$

where $c_{g,1} > c_{e,min}$. Effectively, this barrier function introduces a two-layer definition of safety. As c_e decreases and approaches $c_{g,1}$, $h_{g,1}$ approaches 0. This is an “unsafe” scenario that may potentially be tolerable for brief periods of time. As c_e decreases further and approaches $c_{e,min}$, $h_{g,1}$ approaches -1 . This represents a “catastrophe.” When used with the following nonlinear second-order constraint

$$\ddot{h}_{g,1} + 2\omega\zeta\dot{h}_{g,1} + \omega^2 \frac{h_{g,1}}{h_{g,1} + 1} \geq 0 \quad (12)$$

with a damping ratio ζ and natural frequency $\omega > 0$, the resulting behavior resembles a damped oscillator with a nonlinear spring that stiffens aggressively near $h_{g,1} = -1$ and becomes a less aggressive second-order ZCBF near $h_{g,1} = 0$. In fact, if $h_{g,1}$ is linearized around 0, the constraint approximates a more classical second-order exponential control barrier constraint (Moon, 2025).

Interestingly, this structure enables the graceful CBF to facilitate both induction and maintenance. Since it acts similarly to a classical exponential barrier function in the neighborhood of $c_e = c_{g,1}$, $c_{g,1}$ is chosen to be the desired effect site propofol concentration corresponding to 50% BIS. On the other hand, the singularity in the spring-like term at $h_g = -1$ triggers an increasingly aggressive dosage response which resembles induction. Therefore $c_{e,min}$ in this case is set to be near 0, or $-\delta \mu\text{g mL}^{-1}$, where $\delta = 0.01 \ll 1$. This slight offset is to avoid the singularity for un-drugged trajectories which begin with exactly 0 effect site propofol concentration.

The above measures prevent underdosage. To prevent overdosage, a second graceful CBF is added:

$$h_{g,2}(c_e) := \frac{c_{g,2} - c_e}{c_{e,max} - c_{g,2}} \quad (13)$$

where $c_{g,2} < c_{e,max}$. Here, $c_{e,max}$ is set to the effect site concentration corresponding to 40% BIS. As c_e increases and approaches $c_{g,2}$, $h_{g,2}$ approaches 0. As c_e increases and approaches $c_{e,max}$, $h_{g,2}$ approaches -1 . Again, on the side of overdosage, given a reasonable choice of $c_{g,2}$, there is a “safe” regime where $h_{g,2} > 0$, an “unsafe but not catastrophic” regime where $-1 < h_{g,2} < 0$, and a “catastrophic” regime where $h_{g,2} < -1$. Applying the same nonlinear second-order constraint results in the same behavior near higher dosages of propofol:

$$\ddot{h}_{g,2} + 2\omega\zeta\dot{h}_{g,2} + \omega^2 \frac{h_{g,2}}{h_{g,2} + 1} \geq 0 \quad (14)$$

where ζ and ω are the same as those for $h_{g,1}$ for simplicity. Below is a visual representation of this graceful controller’s four barriers, with corresponding effect site concentrations and BIS levels.

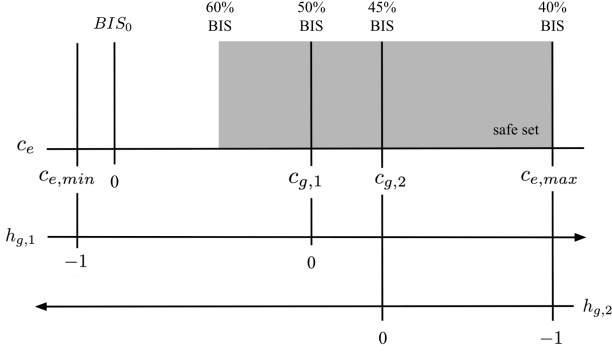


Fig. 1. A visualization of the graceful control barriers.

Having defined these graceful CBFs, it possible to design a safety controller with the following quadratic program:

$$\begin{cases} \text{minimize} & \frac{1}{2}u(t)^2 + \frac{1}{2}p_1\epsilon_1^2 + \frac{1}{2}p_2\epsilon_2^2 \\ \text{subject to} & u(t) + \epsilon_1 \geq u_{min}(t) \\ & u(t) - \epsilon_2 \leq u_{max}(t) \\ & 0 \leq u(t) \leq u_{cap} \end{cases} \quad (15)$$

where ϵ_1, ϵ_2 are slack variables added to ensure feasibility in the presence of control input constraints, with Pareto weights $p_1 = 1 \times 10^8$, $p_2 = 1$. The input bounds $u_{min}(t)$ and $u_{max}(t)$ are computed by plugging the model equations into and rearranging the two graceful safety constraints:

$$\begin{aligned} u_{min}(t) = & V_1(k_{10}c_1 - k_{12}(c_2 - c_1) - k_{13}(c_3 - c_1) + \\ & k_{e0}(c_1 - c_e)) - 2V_1\omega\zeta \frac{c_{g,1} - c_{e,min}}{k_{e0}} \dot{h}_{g,1} - \\ & V_1\omega^2 \frac{c_{g,1} - c_{e,min}}{k_{e0}} \ddot{h}_{g,1} \\ u_{max}(t) = & V_1(k_{e0}(c_1 - c_e) - k_{12}(c_2 - c_1) - \\ & k_{13}(c_3 - c_1) + k_{10}c_1) + 2V_1\omega\zeta \frac{c_{e,max} - c_{g,2}}{k_{e0}} \dot{h}_{g,2} + \\ & V_1\omega^2 \frac{c_{e,max} - c_{g,2}}{k_{e0}} \ddot{h}_{g,2} \end{aligned} \quad (16)$$

We simulated the above controller for different initial conditions that simulate patients beginning in an awake state, patients dangerously close to awakening, and patients who are somewhat over-dosed. As shown in the phase plot in Fig. 2, our proposed controller succeeds in converging to the safe zone for all the initial conditions while avoiding “catastrophic” over- and under-dosage. This reflects the “multi-layered” nature of the safety guarantees provided by graceful control.

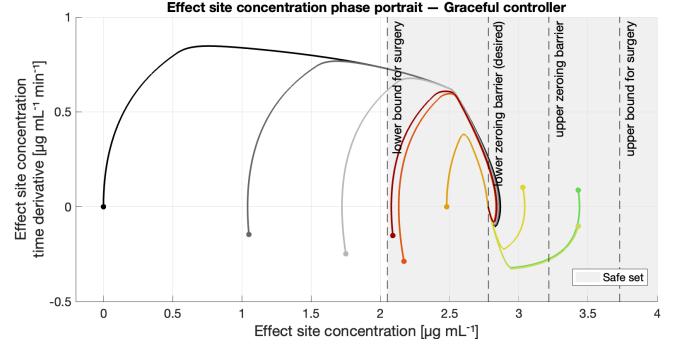


Fig. 2. Graceful controller phase portrait

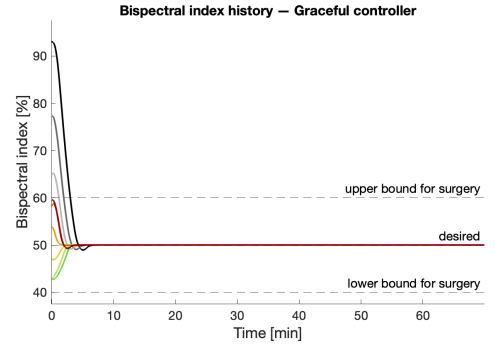


Fig. 3. Graceful controller BIS trajectories

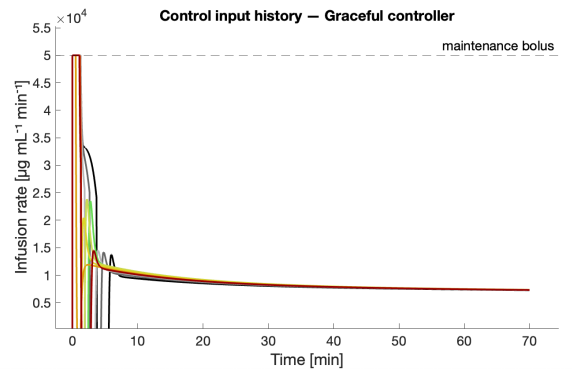


Fig. 4. Graceful controller input trajectories

Figures 3 and 4 show the trajectories of BIS and propofol infusion rate needed for achieving the above graceful safety results. The proposed controller succeeds in achieving both the induction and maintenance of the desired range of BIS values, for all initial conditions simulated. Moreover, this is achieved with a control input trajectory that remains within the feasible bounds, with those bounds being active for only brief fractions of total time.

4.2 Graceful Control with Interval Estimation

The above simulation results assume that all four state variables, c_1, c_2, c_3, c_e , are measured directly. In clinical practice, typically only BIS is measured, and therefore only $c_e(t)$ can be inferred through the potentially inaccurate inversion of the Hill equation. To address this issue, we construct a new anesthesia safety controller that assumes that only the upper and lower bounds on the above four state variables are available. We obtain these upper and lower bounds using the proposed interval estimation algorithm.

The gain matrix $\mathbf{L}$ of the interval observer was designed by placing the poles of $\mathbf{A} - \mathbf{L}\mathbf{C}$ at locations chosen by a particle swarm algorithm that minimizes the worst-case interval width: a problem that can be examined in future research using Linear Matrix Inequality (LMI) methods. Then, a similarity transform $\mathbf{T}$ built from the eigenvectors of $\mathbf{A} - \mathbf{L}\mathbf{C}$ was applied to diagonalize the closed-loop error dynamics. Because a diagonal matrix is trivially Metzler, this guarantees a valid interval observer. The resulting eigenvalues were -5.0000 , -0.7687 , -0.0690 , and -0.0041 . As seen in Fig. 8, this results in very slow convergence for the slow peripherals.

Given the resulting estimator, the new control input bounds $u'_{min}(t)$ and $u'_{max}(t)$ are computed conservatively such that:

$$u'_{min}(t) = \max_{\mathbf{x}(t) \in [\underline{\mathbf{x}}(t), \bar{\mathbf{x}}(t)]} u_{min}(t)$$

$$u'_{max}(t) = \min_{\mathbf{x}(t) \in [\underline{\mathbf{x}}(t), \bar{\mathbf{x}}(t)]} u_{max}(t).$$

The Pareto weights for slack variables remain the same as in the previous simulation study. Feeding the interval observer estimates into the graceful controller in this manner, the resulting phase portrait is shown below. At all moments in time, the output effect site concentration c_e was “measured” by inverting the Hill function given a BIS reading output from the plant, which involves noise. The observer parameters V was set to $0.5 \mu\text{g mL}^{-1}$, $\bar{\mathbf{d}}(t)$ was set to a constant $0.03 \mu\text{g mL}^{-1}$ in each component, and $\underline{\mathbf{d}}(t)$ was set to a constant $-0.03 \mu\text{g mL}^{-1}$ in each component. The initial interval estimates in each simulation were set to depart by $+0.3 \mu\text{g mL}^{-1}$ and $-0.3 \mu\text{g mL}^{-1}$ for each state from the true initial states.

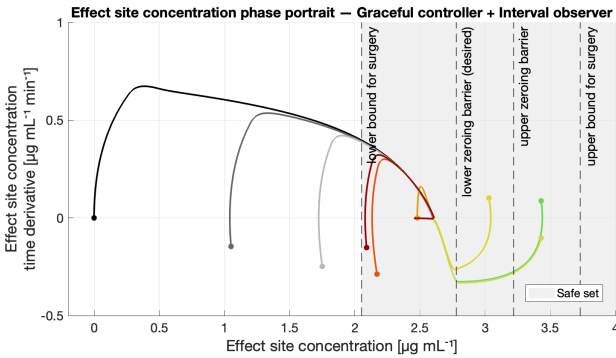


Fig. 5. Graceful control phase plot with interval estimation

As shown in Figures 5-7, combining the proposed graceful controller with interval estimation continues to guarantee

patient safety for a broad range of initial conditions. Furthermore, as shown in Fig. 8, the interval observer succeeds in its mission - namely, to provide upper and lower estimation bounds that “sandwich” the true state trajectories. This is an encouraging set of results demonstrating the viability of the proposed algorithms.

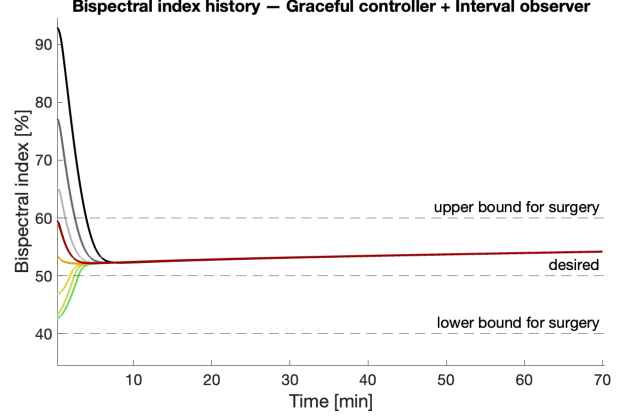


Fig. 6. Graceful control BIS plots with interval estimation

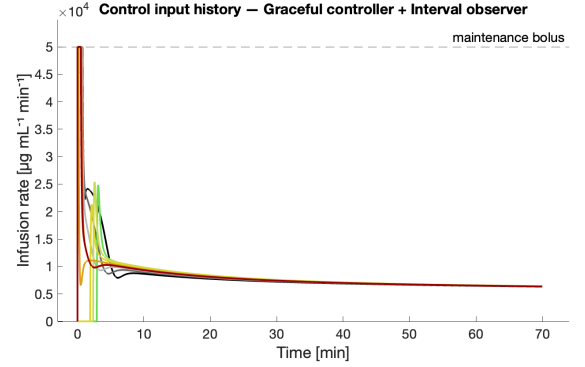


Fig. 7. Graceful control inputs with interval estimation

5. CONCLUSION

The main conclusion of this paper is that graceful safety control, combined with interval estimation, can provide multi-layered safety guarantees for automated propofol sedation. As our controller was simulated on only one patient model, potential topics for future research include simulation under modeling uncertainties to account for patient-to-patient variability across demographics.

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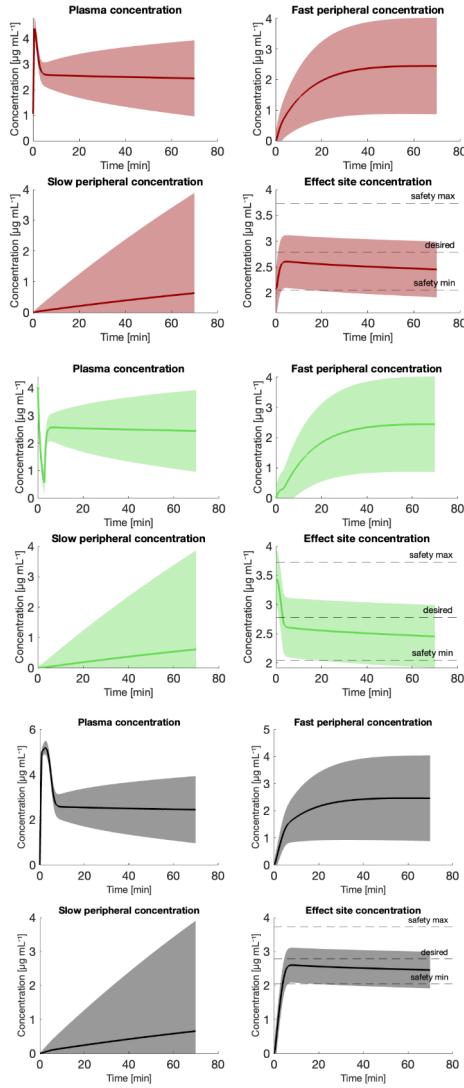


Fig. 8. Interval estimation trajectories

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